allantis v1

2026-05-02 15:57:33

Research Question

Which entry-time IV conditions predict my best and worst trades?

Mode: Backtest Trade Regime Analysis (agentic)

Date range: 2021-01-15 to 2026-04-10

Model: claude-sonnet-4-6

Workflow Plan

Task Type: backtest-regime-analysis

Reasoning: Backtest IV analysis: 248 trades, 244 matched to entry-morning IV (98% match rate). Claude will explore 84 IV metric columns vs trade outcomes.

Hypotheses:

• Some entry-time IV metrics predict final trade P&L

• IV regime at entry conditions win/loss probability

• The IV predictive signal is stable across different time periods

Feature columns (X): spot, vix_1d, vix_7d, vix_30d, vix_90d, vix_180d, iv_1d_25c, iv_1d_25p, iv_1d_atm, iv_7d_25c, iv_7d_25p, iv_7d_atm, forward_1d, forward_7d, iv_30d_25c, iv_30d_25p, iv_30d_atm, iv_90d_25c, iv_90d_25p, iv_90d_atm, forward_30d, forward_90d, iv_180d_25c, iv_180d_25p, iv_180d_atm, forward_180d, skew_1d_10p_25p, skew_1d_10p_atm, skew_1d_25p_25c, skew_1d_25p_atm, skew_1d_atm_10c, skew_1d_atm_25c, skew_7d_10p_25p, skew_7d_10p_atm, skew_7d_25p_25c, skew_7d_25p_atm, skew_7d_atm_10c, skew_7d_atm_25c, skew_30d_10p_25p, skew_30d_10p_atm, skew_30d_25p_25c, skew_30d_25p_atm, skew_30d_atm_10c, skew_30d_atm_25c, skew_90d_10p_25p, skew_90d_10p_atm, skew_90d_25p_25c, skew_90d_25p_atm, skew_90d_atm_10c, skew_90d_atm_25c, term_ratio_1d_7d, skew_180d_10p_25p, skew_180d_10p_atm, skew_180d_25p_25c, skew_180d_25p_atm, skew_180d_atm_10c, skew_180d_atm_25c, term_ratio_7d_30d, term_ratio_30d_90d, term_slope_1_7_25c, term_slope_1_7_25p, term_slope_1_7_atm, term_slope_7_30_25c, term_slope_7_30_25p, term_slope_7_30_atm, days_to_monthly_opex, term_slope_30_90_25c, term_slope_30_90_25p, term_slope_30_90_atm, convex_1d_10p_25p_atm, convex_1d_25p_atm_25c, convex_1d_atm_25c_10c, convex_7d_10p_25p_atm, convex_7d_25p_atm_25c, convex_7d_atm_25c_10c, convex_30d_10p_25p_atm, convex_30d_25p_atm_25c, convex_30d_atm_25c_10c, convex_90d_10p_25p_atm, convex_90d_25p_atm_25c, convex_90d_atm_25c_10c, convex_180d_10p_25p_atm, convex_180d_25p_atm_25c, convex_180d_atm_25c_10c

Outcome columns (Y): pnl, is_win

Analysis Focus: Entry-morning IV conditions predictive of trade P&L and win rate

Research Report

This analysis examined 244 IV-matched trades from the Allantis strategy (Jan 2021 - Apr 2026) to identify which entry-morning IV surface conditions best predict final trade P&L and win rate. Two dominant, economically coherent signals emerge: (1) **medium-term put skew** (90d and 180d tenor), where more elevated put-side skew at entry reliably predicts worse outcomes, and (2) **call-side convexity** (90d and 30d), where higher call-wing curvature at entry predicts better win rates. These are entry-time predictors -- IV conditions captured at 09:35 on the entry date -- not post-trade or coincidental correlations.

The skew signal (skew_90d_25p_25c, r = –0.25; Spearman r = –0.31) shows a clear monotonic P&L degradation as puts become more expensive relative to calls: top-decile skew (least negative) trades had a mean P&L of just $74 and a 41.7% win rate, versus $1,566 mean P&L and 95.8% win rate in the most-favorable decile. The convexity signal (convex_90d_atm_25c_10c) displays a threshold effect: trades entered when call-wing convexity falls into the lowest bucket (≤0.0042) suffer a 56.3% win rate and mean P&L of just $383, while all higher buckets average 75-92% win rates.

Critically, the rolling-correlation analysis reveals that the skew signal is not always "on" -- it was near-zero or even inverted in early 2022, surged to r = –0.52 to –0.62 during 2022-2023 stress periods and re-emerged strongly in 2024-2026, suggesting the signal amplifies in regimes where the market is actively pricing tail risk into the medium-term surface. Position sizing or trade-selection filters conditioned on these two IV metrics could materially improve risk-adjusted returns.

Signal 1: Medium-Term Put Skew (skew_90d_25p_25c and relatives)

The single strongest predictor across both pnl (r = –0.250, p < 0.001, n = 244) and is_win (r = –0.250, p < 0.001) is skew_90d_25p_25c -- the IV spread between 90-day 25-delta puts and 25-delta calls. The Spearman rank correlation is even stronger (r = –0.308), confirming non-linearity. The decile profile is stark: the two most favorable deciles (deeply negative skew, cheapest puts relative to calls) show win rates of 88-96% and mean P&Ls of $1,158-$1,566, while the worst decile (least negative skew, most expensive puts) collapses to a 41.7% win rate and mean P&L of just $74. The tail analysis confirms the top-10% P&L trades entered with skew_90d_25p_25c averaging –0.340 vs. –0.294 for the bottom-10% trades -- a 0.046 difference that is economically large relative to the full range.

This pattern is consistent across tenor (180d and 30d versions all rank in the top-15), and holds in both high- and low-VIX regimes: the vix_30d median-split shows r = –0.266 (high-VIX) vs. r = –0.232 (low-VIX), confirming it is not purely a VIX-level artifact.

Signal 2: Call-Wing Convexity (convex_90d_atm_25c_10c and convex_30d_atm_25c_10c)

The convexity metrics measure curvature of the call-side of the surface (from ATM to 25-delta to 10-delta call). These carry r = +0.170 to +0.226 vs. is_win -- the opposite sign from skew, meaning that a fatter, more convex call wing at entry is associated with better outcomes. The win-rate buckets for convex_90d_atm_25c_10c reveal a critical threshold: bucket 1 (values 0.0021-0.0042) has just a 56.3% win rate and $383 mean P&L. Buckets 2-5 all exceed 75% win rate with means of $976-$1,345. The convex_30d_atm_25c_10c confirms this: win rate rises monotonically from 62.5% (lowest bucket) to 92.3% (highest bucket), a spread of 29.8 percentage points.

Regime and Time-Stability Analysis

The rolling 40-trade correlation for skew_90d_25p_25c shows significant time-variation. The signal was strong at –0.39 in early 2021, went near-zero or positive through mid-2022 (likely as the 2022 bear market caused unusual put/call dynamics), then re-intensified to –0.53 to –0.62 in late 2022 through 2023, and is currently running at approximately –0.50 to –0.62 in 2024-2026. The 3-period time-split confirms skew is a top signal in both Period 2 (r = –0.44 to –0.47) and Period 3 (r = –0.29 to –0.32), though less prominent in Period 1 (where convexity and 10p-25p skew were the dominant signals). Convex_7d_atm_25c_10c appears as the #3 signal in Period 3 (r = +0.31), reinforcing the convexity theme across recent data.

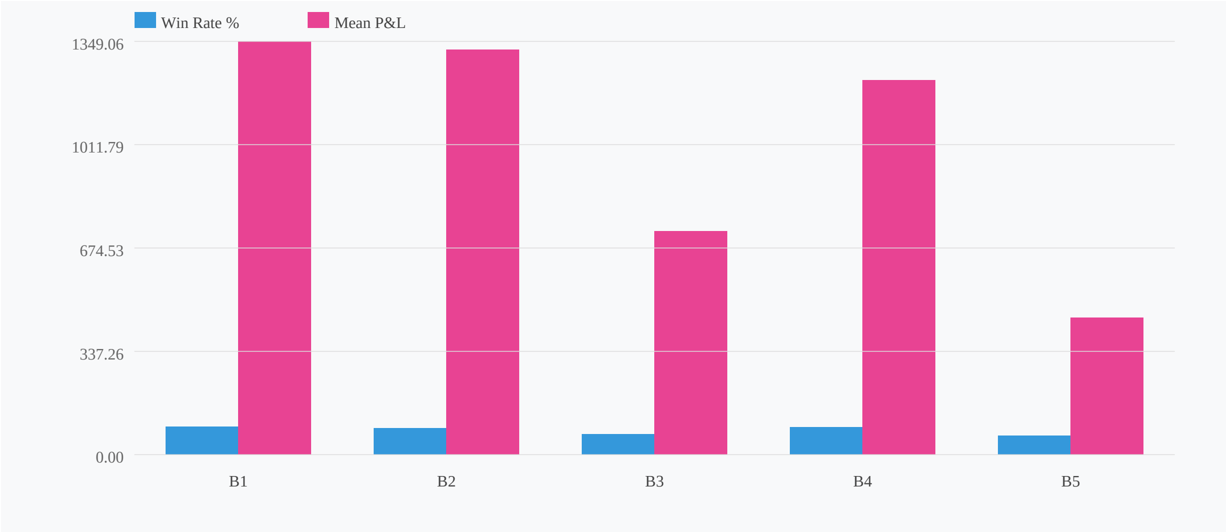
Combined Regression

A four-variable OLS (skew_90d_25p_25c, skew_180d_25p_atm, convex_90d_atm_25c_10c, vix_90d) yields $R^2 = 0.085$ -- modest but statistically meaningful for a binary-outcome trade dataset. VIX_90d contributes positively (coef = +6,149), consistent with premium-selling strategies benefiting from elevated IV at entry. The negative convex_90d coefficient in OLS (–81,474) reflects collinearity within the combined model, not a reversal of its win-rate relationship -- the win-rate bucket analysis is the more reliable view here.

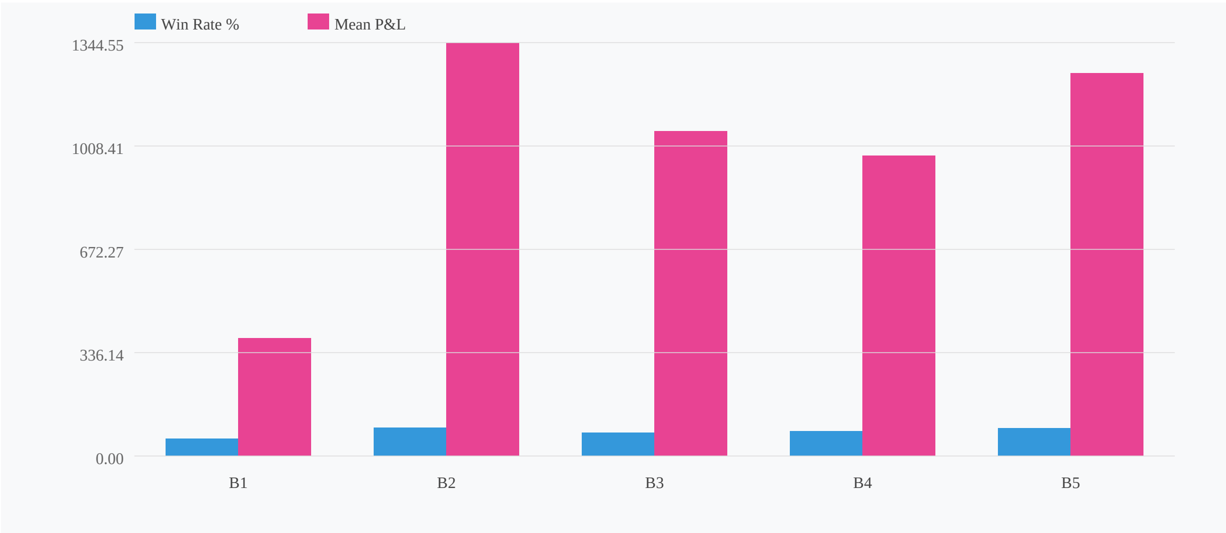
What This Means Structurally

The put skew signal likely reflects a market-fear premium: when the 90d surface is pricing unusually expensive downside protection relative to upside, this strategy tends to underperform. This could indicate the strategy has positive delta/negative vega characteristics that suffer when the market is in a "protected but nervous" state. The call convexity signal may capture the opposite -- when the call wing is well-shaped (convex), there is more structural premium available on the call side to harvest, or the surface is in a more "normal" configuration.

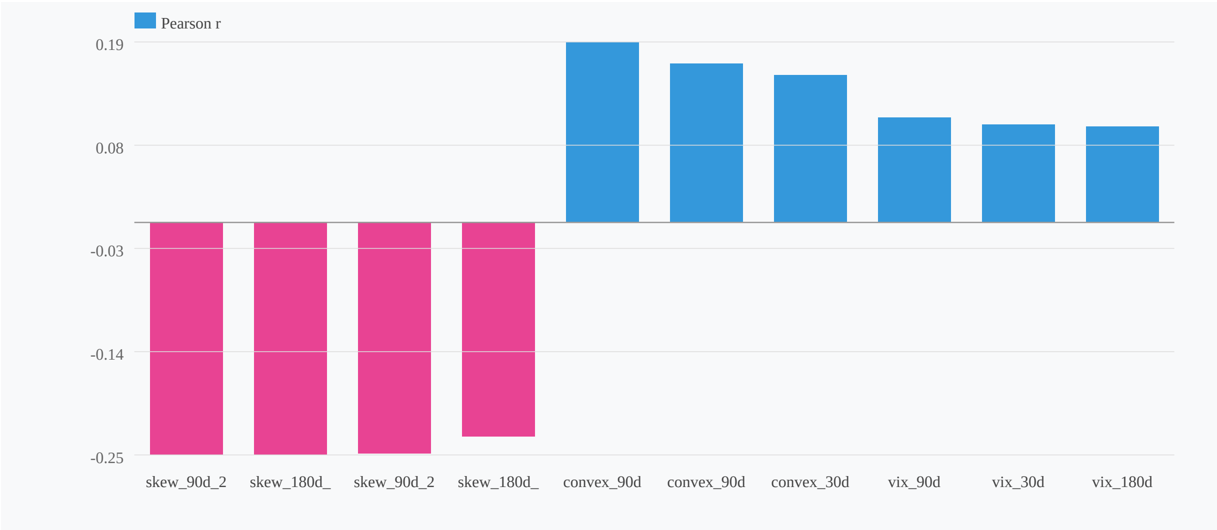
Win Rate & Mean P&L by skew_90d_25p_25c Bucket



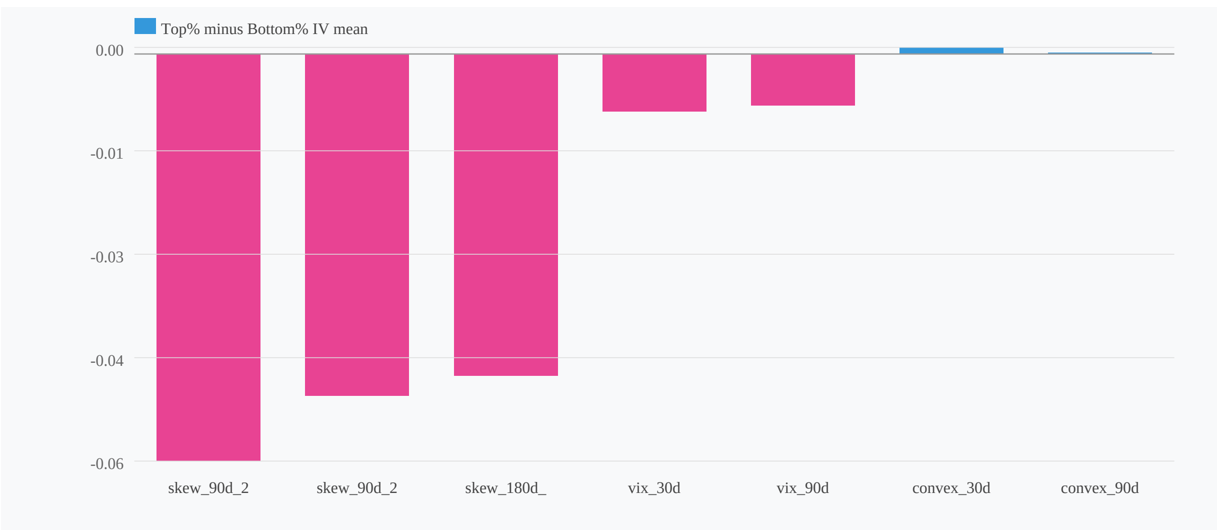
Win Rate & Mean P&L by convex_90d_atm_25c_10c Bucket



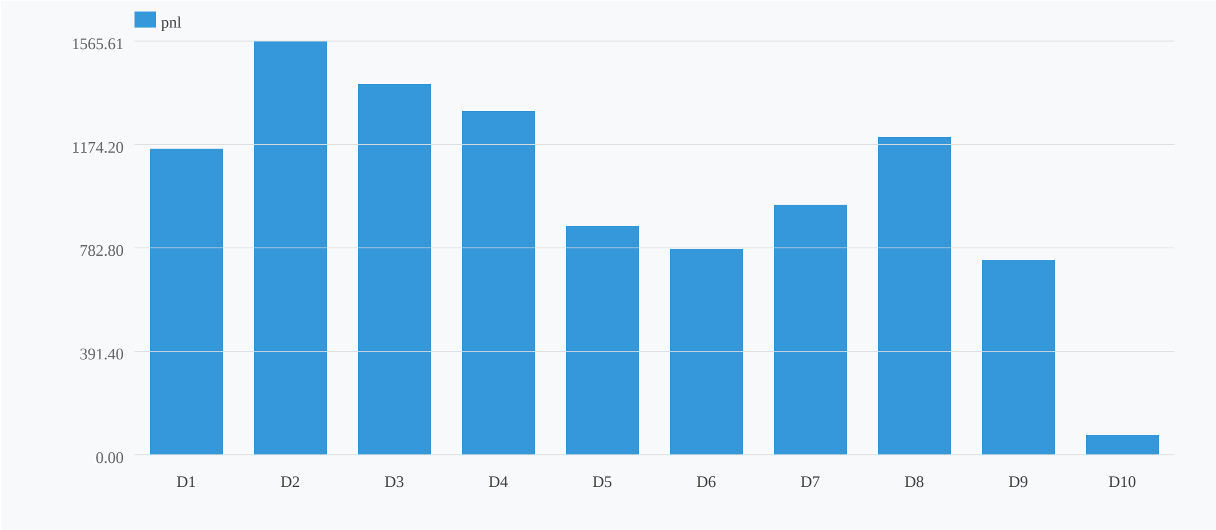
IV Correlation with pnl (top 10)



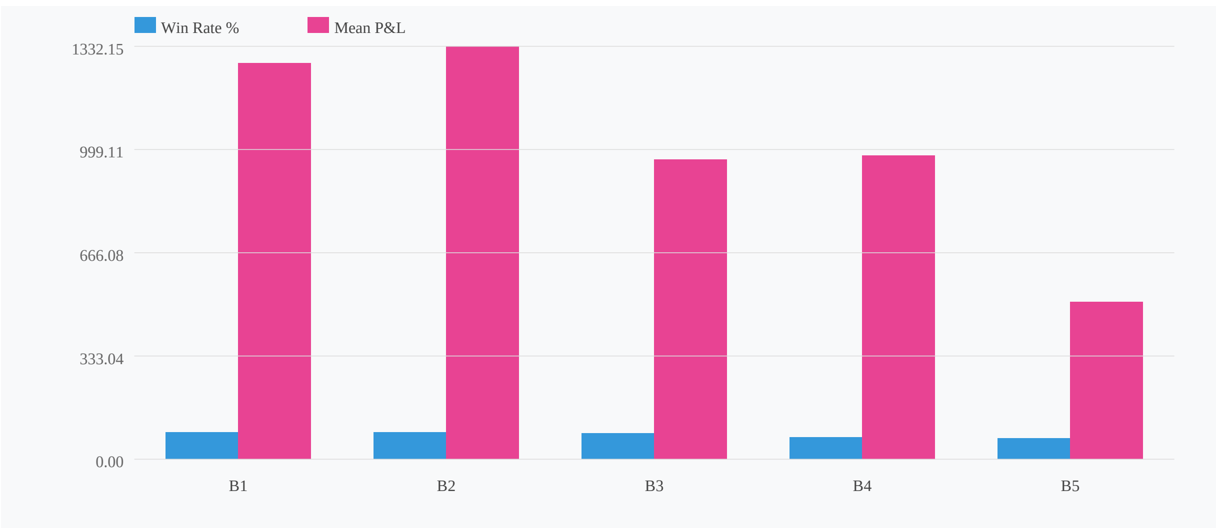
IV Difference: Best vs Worst 10% pnl



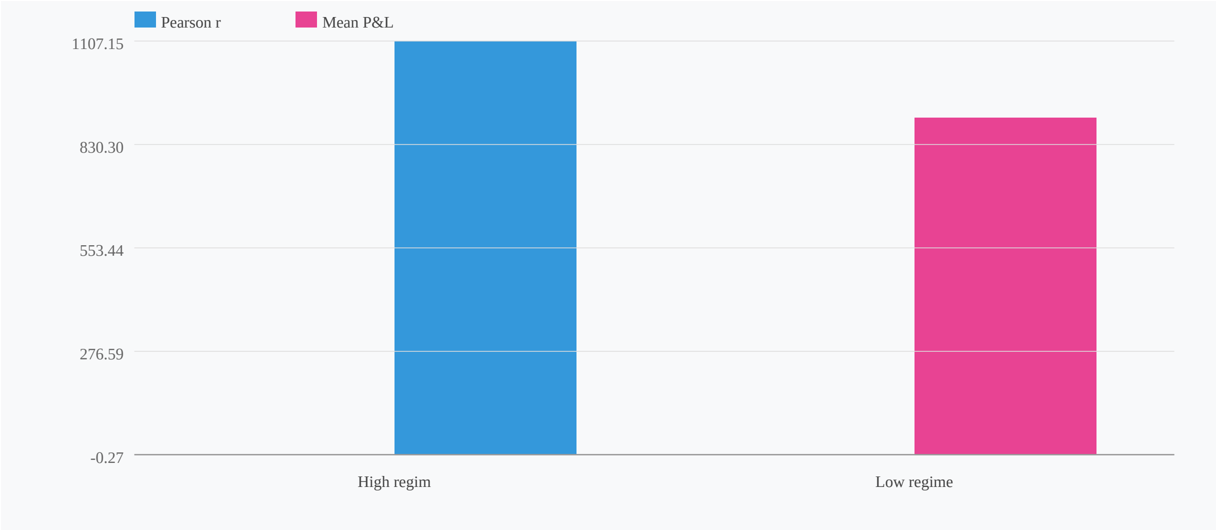
Decile Profile: skew_90d_25p_25c → pnl (r=-0.250)



Win Rate & Mean P&L by skew_90d_25p_atm Bucket



Regime Split: skew_90d_25p_25c by vix_30d



Correlation Stability Across Periods (consistency=0%)